

Defining the Forward measure

Define a new measure: $\frac{d\mathbb{Q}^T}{d\mathbb{Q}} := \frac{1}{P(0, T)B(T)} = \mathcal{E}_T(\nu(\cdot, T) \bullet W^*) \quad \nu(t, T) = - \int_t^T \sigma(t, u) du$

$$\frac{d\mathbb{Q}^T}{d\mathbb{Q}} \Big|_{\mathcal{F}_t} = \mathbb{E}_t^{\mathbb{Q}} \left[\frac{P(T, T)}{P(0, T)B(T)} \right] = \frac{P(t, T)}{P(0, T)B(t)}$$

Idea: Under this new measure, we can simplify some calculations (New Numeraire).

$$\frac{d\mathbb{Q}}{d\mathbb{Q}^S} = \frac{B(S)P(0, S)}{P(S, S)} \quad \frac{d\mathbb{Q}^S}{d\mathbb{Q}^T} \Big|_{\mathcal{F}_t} = \frac{P(t, S)P(0, T)}{P(t, T)P(0, S)} = \mathcal{E}_t(\sigma_{S, T} \bullet W^T)$$

Use Bayes rule for measure change calculations: $\mathbb{E}^{\mathbb{Q}}[X|\mathcal{F}] = \frac{\mathbb{E}^{\mathbb{P}}[X \frac{d\mathbb{Q}}{d\mathbb{P}}|\mathcal{F}]}{\mathbb{E}^{\mathbb{P}}[\frac{d\mathbb{Q}}{d\mathbb{P}}|\mathcal{F}]}$

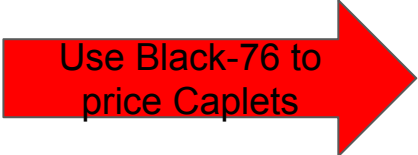
Then we can find that $B(s)/P(s, T)$ is a martingale under $\mathbb{Q}^T$

$$\text{Main results: } \pi(t) = B(t)\mathbb{E}_t^{\mathbb{Q}} \left[\frac{X}{B(T)} \right] = P(t, T)\mathbb{E}_t^{\mathbb{Q}^T}[X] \quad \mathbb{E}_t^{\mathbb{Q}^T}[r(T)] = \mathbb{E}_t^{\mathbb{Q}^T}[f(T, T)] = f(t, T)$$

The Libor Market Model (LMM)

Assuming deterministic $\sigma \rightarrow$ ZCP-prices Log-normal $\rightarrow$ Spot rate cannot be

$$F(T, T + \delta) = \frac{1}{\delta} \left(\frac{1}{P(T, T + \delta)} - 1 \right) \quad \text{Fix: Model rates directly}$$

Old way: $F(T, T + \delta) \sim e^x$  Use Black-76 to price Caplets $Cpl(0) = P(0, T + \delta) \text{Black}(F(0, T, T + \delta), K, T, \sigma)$

HJM way: $dL(t, T) = L(t, T)\lambda(t, T)dW^{T+\delta}$
(Deterministic λ)

$$\sigma(t, T + \delta) = \sigma(t, T) + \partial_T \lambda(t, T)(1 - e^{-\int_T^{T+\delta} f(t, u) du}) + \lambda(t, T)(f(t, T + \delta) - f(t, T))e^{-\int_T^{T+\delta} f(t, u) du}$$

Discrete way: $dL(t, T_{M-1}) = L(t, T_{M-1})\lambda(t, T_{M-1})dW^{T_M}(t)$

$$\frac{dL(t, T_m)}{L(t, T_m)} = \begin{cases} -\lambda(t, T_m) \sum_{l=m+1}^n \lambda(t, T_l)^\top \frac{\delta L(t, T_l)}{\delta L(t, T_l) + 1} dt + \lambda(t, T_m) dW^{T_{n+1}} & m < n \\ \lambda(t, T_m) dW^{T_{m+1}} & m = n \\ +\lambda(t, T_m) \sum_{l=n+1}^m \lambda(t, T_l)^\top \frac{\delta L(t, T_l)}{\delta L(t, T_l) + 1} dt + \lambda(t, T_m) dW^{T_{n+1}} & m > n \end{cases}$$

There exists a measure where L is log-normal! $\rightarrow$ Still hard to price multirate.

Negative interest rates

We want the model to reproduce negative interest rates

Fix 1: 'Move' the base-line. Add a constant $\alpha > 0$ such that $X(t, T) = L(t, T) + \alpha$ is the log-normal variable we want to model.

$$\frac{dL(t, T_m)}{L(t, T_m) + \alpha} = \begin{cases} -\lambda(t, T_m) \sum_{l=m+1}^n \lambda(t, T_l)^\top \frac{\delta(L(t, T_l) + \alpha)}{\delta L(t, T_l) + 1} dt + \lambda(t, T_m) dW^{T_{n+1}} & m < n \\ \lambda(t, T_m) dW^{T_{n+1}} & n = m \\ +\lambda(t, T_m) \sum_{l=n+1}^m \lambda(t, T_l)^\top \frac{\delta(L(t, T_l) + \alpha)}{\delta L(t, T_l) + 1} dt + \lambda(t, T_m) dW^{T_{n+1}} & m > n \end{cases}$$

Careful: Large α creates large volatility smiles.

Fix 2: Use Bachelier model: $dL(t, T_{M-1}) = \lambda(t, T_{M-1}) dW^{T_M}(t)$

Eliminates state dependence

Monte Carlo and the Spot measure (LMM)

Reduce simulation error by using the spot measure:

$$\frac{dQ^*}{dQ^{T_M}} \Big|_{\mathcal{F}_t} = \mathcal{E}_t \left(\sigma_{T_0, T_{M-1}} \bullet W^{T_M} \right)$$

$$\sigma_{T_0, T_{M-1}}(t) = \sum_{k=\eta(t)}^{M-1} \sigma_{T_k, T_{k+1}}(t)$$

$$B^*(0) = 1$$

$$B^*(T_m) = (1 + \delta L(T_{m-1}, T_{m-1})) B^*(T_{m-1})$$

$$\mathbb{E}_{T_m}^{Q^{T_M}} \left[\frac{B^*(T_M)}{P(T_M, T_M)} \right] = \frac{B^*(T_m)}{P(T_m, T_M)}$$

$$\frac{dL(t, T_m)}{L(t, T_m)} = \lambda(t, T_m) \sum_{k=\eta(t)}^m \lambda(t, T_k)^\top \frac{\delta L(t, T_k)}{\delta L(t, T_k) + 1} dt + \lambda(t, T_m) dW^*(t)$$

Swaption and the Swap measure

Swaption → Option to enter into an interest rate swap at the first payment date.
Essentially a weighted average of rates.

$$\begin{aligned}\pi(t) &= P(t, T_\mu) \mathbb{E}_t^{\mathbb{Q}^{T_\mu}} [(\Pi(T_\mu))^+] \\ &= P(t, T_\mu) \mathbb{E}_t^{\mathbb{Q}^{T_\mu}} [A(T_\mu)(R_{\text{swap}}(T_\mu) - K))^+] \end{aligned}$$

$$A(t) := \sum_{i=\mu+1}^{\nu} \delta P(t, T_i)$$

$$R_{\text{swap}}(t) = \frac{P(t, T_\mu) - P(t, T_\nu)}{A(t)}$$

Swap measure:

$$\frac{d\mathbb{Q}^{\text{swap}}}{d\mathbb{Q}^{T_\mu}} = \frac{P(0, T_\mu)}{A(0)} \frac{A(T_\mu)}{P(T_\mu, T_\mu)}$$

Essentially use $A(t)$ as the numeraire

Under log-normal LMM: $R_{\text{swap}}(t) = \frac{P(t, T_\mu) - P(t, T_\nu)}{A(t)} = \sum_{i=\mu+1}^{\nu} w_i(t) L(t, T_{i-1}), \quad w_i(t) = \frac{\delta P(t, T_i)}{\sum_{j=\mu+1}^{\nu} \delta P(t, T_j)}$

Under the swap measure, R is a martingale, and we just choose dynamics to make it log-normal:

$$\frac{dR_{\text{swap}}(t)}{R_{\text{swap}}(t)} = \rho(t) dW^{\text{swap}}(t)$$